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ECMT 461 - 904

Final Term Project

The Relationship between Health Insurance Coverage and Age Among Different Regions

I: Introduction

How does the likelihood of being covered by health insurance vary among different age groups across different regions in the United States and what could be the key factors that would contribute to this likelihood? I think this is an important question to ask because of the many implications the answer to this question could have for the health of many different people (even those outside of the U.S). Also trying to analyze the answer to this question should give us a better idea on the people that are most likely to be at most risk when it comes to dealing with health related problems, but also what could be the potential factors that are causing this risk in the first place. Having an understanding of this could also make us better understand the effects of public policy related to health insurance as well and how much they are really helping.

I do expect some differences between regional health insurance coverage rates due to potential differences in average income for each of the U.S regions (Northeast, Midwest, South, West). However even though I am expecting some differences between health insurance coverage rates between U.S regions, I expect for there to be larger differences between coverage rates among different age groups. In particular, I expect that those under the age of 18 would have a relatively high likelihood of being covered by health insurance possibly because of their parents providing that coverage for them. I would expect that from around the 18-35 age group,

the likelihood of someone being covered by health insurance actually decreases. Several reasons for this that I could think of are the likelihood that many people within this age group would consider themselves low health risk and would go without insurance as a method to save money. Essentially this would be an example of adverse selection in the health insurance market. Also there is the likelihood that many people in this age group are likely to be in the earlier stages of their careers and are less likely to have money available to purchase health insurance and so in turn they would choose not to. For someone that is in the 35-65 age group, the likelihood that they would be covered by health insurance would likely increase. The primary reason for this is that many people in this age group would likely have more experience in their careers at this point and so they would likely be earning more than they were previously. Also as people get older, they are more likely to become higher health risk than they were previously giving this age group more incentive to purchase insurance that they are also more likely to be able to afford at this point. As far as what I expect for the overall insurance coverage rate for the entire nonelderly working age group, I would expect it to be comparable to the insurance coverage rate for the under 18 age group. Lastly for the age group that is 65 years and older, the likelihood that they will be covered by health insurance is very high. First reason I can think of for this is that many people in this age group would likely consider themselves relatively high health risk and so would have the most incentive to get health insurance. Also in the U.S, Medicare provides health coverage for those over the age of 65 regardless of their income. Overall, I do feel that adverse selection in the health insurance market (where people who consider themselves higher health risk are more likely to purchase insurance than those who would consider themselves low health risk) is likely to play a role in how much health insurance coverage different age groups have.

II: Literature Review

After taking a look at various publications that analyzed the issue of health insurance coverage, I began to notice consistent trends in what these publications found. Across the board, it had been concluded that the cost of health insurance had been the primary factor for why many people went without it. While the publications that I read did mention similar findings, they also would often analyze different factors which can help one get a fuller picture of what is causing people to be uninsured. In the publication “Key Facts about the Uninsured Population” by The Henry J. Kaiser Family Foundation, it was noted that the reason why many people would not opt for health insurance was because many of those uninsured did not even have a regular place to visit when they needed medical attention or advice. Also it was noted that there were a fair amount of uninsured people that were employed but did not have access to health insurance coverage from their employer. The publication also highlighted an issue that many low income families face where they would earn too much to qualify for governmental assistance programs like Medicaid but not enough to sufficiently afford individual health insurance. It is these families that often find themselves choosing to take the risk of being uninsured. In a publication from the United States Census Bureau “Who Are the Uninsured?” by Edward Berchick, it was pointed out how education had a role to play in whether someone would have coverage. In particular, people who had a high school education or less were more likely to be uninsured. In another publication called “Who Goes Without Health Insurance? Who Is Most Likely to Be Uninsured?” by the Institute of Medicine (U.S), it was noted how overall the Southern and Western regions of the U.S had the highest uninsured rates regardless of income. The publication stated that there is not a very clear answer as to why the uninsured rates are higher in the Southern and Western regions of the U.S, but that there are likely multiple different factors that

are causing this difference. The factors the publication particularly mentioned were population characteristics (racial and ethnic group composition), employment, and geographic variations. All three of the publications noted that most of those uninsured were nonelderly working age adults (age 19-64). The information and key findings that I gathered from these publications are likely to reappear in my own analysis of this subject in some form. However, what I intend to look more into with my analysis is if differences exist between regional uninsured rates among different age groups and what factors could potentially cause these differences.

III: Methodology/Data

When it came to my statistical analysis of this subject, I have so far used three sets of data in particular. The first set of data that I used is from the U.S Census Bureau titled “Table HI01. Health Insurance Coverage Status and Type of Coverage by Selected Characteristics: 2017”. The total sample size for this data is 323,156 with 73,963 being in the under 18 age group, 198,113 being in the nonelderly adult group (age 18-64), and 51,080 being in the 65 years and older age group. The second set of data that I used is also from the U.S Census Bureau and is titled “PINC-01. Selected Characteristics of People 15 Years Old and Over by Total Money Income in 2017, Work Experience in 2017, Race, Hispanic Origin, and Sex”. The total sample size for this set of data is 262,172 with 42,341 being in the 15 to 24 age group, 168,751 in the 25 to 64 age group, and 51,080 in the 65 years and older age group. The last data set that I used from the U.S Census Bureau is titled “Table HI05. Health Insurance Coverage Status and Type of Coverage by State and Age for All People: 2017”. The total sample size for this set of data is 320,775 with 78,075 being in the Under 19 age group, 193,199 in the 19 to 64 age group, and 49,500 in the 65 years and older age group. I primarily used the first two data sets to observe

health insurance coverage trends that were relevant to this subject. I used the third data set for my actual statistical hypothesis tests.

When it comes to each of the 4 U.S regions, there do appear to be differences in uninsured rates. According to the first data set, the overall uninsured rate for the Northeast region is 5.8% compared to 6.7% for the Midwest region, 11.8% for the Southern region, and 8.2% for the Western region. As can be seen, The Southern and Western regions of the U.S have the highest uninsured rates. What I found here very much correlates with the findings of the Institute of Medicine (U.S) publication that I previously mentioned in the literature review. According to the first set of data, the trend has been that the lower the income, the higher the uninsured rate. However when I refer to the second data set which shows the average incomes of people among each of the regions it shows that the Northeast region has an average income of \$52,462 compared to \$46,698 for the Midwest region, \$45,752 for the Southern region, and \$49,967 for the Western region. Even though the Midwest region had a lower average income in comparison to the Western region, it also had a notably lower uninsured rate. As can be seen, income alone does not seem to explain the differences between uninsured rates between different regions, but it is still likely an important factor.

When it came observing uninsured rates among different age groups, the first aspect that I noticed when I looked at the first set of data is that the uninsured rate is quite low (5.3%) for those under the age of 18. The other trend I noticed was that those under the age of 18 had a higher likelihood of being covered by a government health plan. The likelihood of a child being covered by a government health plan also appeared to be closely linked to being covered by Medicaid. In fact after testing the correlation between the two factors, I found the correlation statistic between the two factors to be .998 which shows that they are indeed likely to be closely

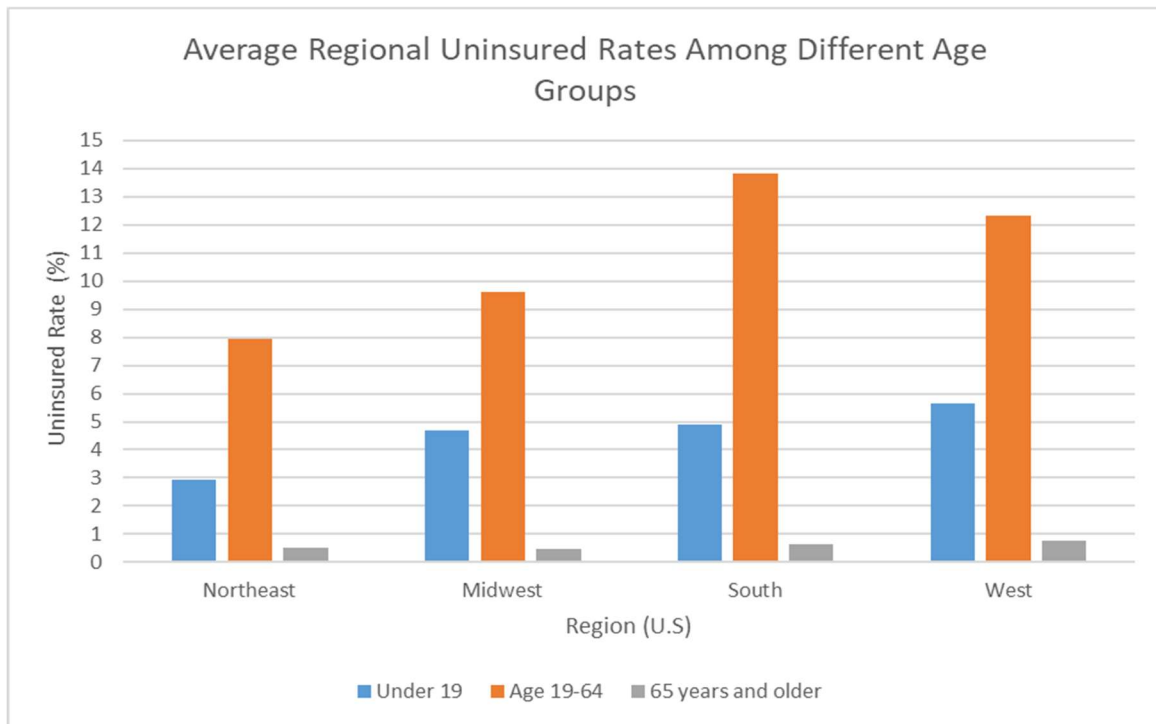
linked. Health insurance provided by the government appears to account for a sizable portion of the health insurance coverage that the under 18 age group receives. This is made more significant considering that people within this age group are less likely to have other kinds of coverage (like private and employment based insurance). In fact while the average rate of private insurance coverage for the 18-64 age group was 72.7%, the average rate for those under 18 was 62.9%. Employment based insurance coverage follows the same pattern, with the average rate of employment based insurance coverage for the 18-64 age group being 63.3% and the average rate for those under 18 being 56.5%. However the story changes when looking at the average coverage rate for government provided health insurance among different age groups. For the 18-64 age group, the average coverage rate for government provided health insurance is 21.8%. On the other hand, the average coverage rate for those under 18 is 42.3%. As can be seen, the under 18 age group has nearly twice the average coverage rate for government provided health insurance in comparison to the nonelderly adult age group (age 18-64). This is a potential explanation for why despite having lower coverage rates for other forms of health insurance, those in the under 18 age group still have an average uninsured rate of 5.3% in comparison to the nonelderly adult age group which has a notably higher average uninsured rate of 12.1%.

Another aspect that I noticed is that the health insurance coverage rate for the nonelderly adult age group (age 18-64) was overall the lowest out of the 3 main age groups that I looked into. Several other statistics that come to mind when I think of this is the difference between the amount of government provided health insurance that those in the nonelderly age group receive in comparison to those who are under 18. As has been stated previously, the average government provided health insurance coverage rate for those under 18 was 42.3% while the average rate for the nonelderly adult group rather sharply dropped to 21.8%. The far lower likelihood of being

provided with government provided health insurance does appear to definitely play a role in the nonelderly adult age group having the highest uninsured rate. What is also interesting to note however is that this age group also has the highest average coverage rates for both private (72.7%) and employment based (63.3%) insurance in comparison to the other 2 age groups. Also statistically speaking according to the first data set, those with higher incomes are more likely to be insured. According to the second set of data that I used, I was able to make note of the incomes that various age groups were making. On average the 15-24 age group made \$18,184, the 25-64 age group made \$56,049, and the 65 years and older age group made \$40,120. As can be seen, the nonelderly adult age group that has the highest uninsured rate actually earns the most. Now I am not claiming that those with higher incomes are more likely to be uninsured, in fact it is the other way around. If all the age groups were to not receive government provided health insurance, then the nonelderly adult age group would most likely be the most insured group holding all else equal.

When it comes to the age group that is 65 years and older, the reason for this age group having the lowest uninsured rate out of all the other age groups in the U.S appears to be quite simple. Simply put, it is because of Medicare. If one is 65 years or older, then Medicare is highly likely the reason why someone in that age group would be covered by health insurance. The statistics here tell it all: this age group has an uninsured rate of 1.3% which is lower than the 5.3% uninsured rate for those under 18 and the 12.1% uninsured rate that nonelderly adults have. Also, Medicare has an average coverage rate of 93.2% for the age group that is 65 years and older versus only 3% for the entire age group that is under 65 years old. Overall, Medicare definitely appears to be the primary reason for why the 65 years and older age group has such a low uninsured rate.

What I intended to test in particular is whether average uninsured rates differ across regions among the three age groups. I used the third set of data which presented data on the health insurance coverage rates among different age groups at the state level to help in achieving this. I made three tables for the three age groups and grouped each of the tables' respective age group's uninsured rates for each of the states into their respective regions. Once I had all the data grouped together, I calculated the average uninsured rate for the different regions among the three age groups. Below is a graphical representation of what I calculated.



Source: "Table HI05. Health Insurance Coverage Status and Type of Coverage by State and Age for All People: 2017" The United States Census Bureau, 2018

I then proceeded to calculate the standard error values for the data I grouped together. Putting all the sample mean, standard error, and sample size values for the grouped data into a table gave the resulting summary table shown below.

Age Group	Statistic	Region			
		Northeast	Midwest	South	West
Under 19 years	Mean	2.9222	4.6833	4.9	5.6462
	Standard Error	0.4017	0.4289	0.5650	0.7073
	Sample Size	9	12	17	13
19 - 64 years	Mean	7.9556	9.6	13.8059	12.3077
	Standard Error	0.7853	0.7465	1.2681	0.9536
	Sample Size	9	12	17	13
65 years and over	Mean	0.5	0.475	0.6471	0.7462
	Standard Error	0.0833	0.0605	0.0986	0.0991
	Sample Size	9	12	17	13

Source: "Table HI05. Health Insurance Coverage Status and Type of Coverage by State and Age for All People: 2017" The United States Census Bureau, 2018

Since there were three age group tables that had to be analyzed, I had to conduct three separate hypothesis tests. For all three age groups, I used the same null and alternative hypothesis. The null hypothesis being that average uninsured rates across all the regions are the same. The alternative hypothesis being the that the average uninsured rates across all the regions are not the same.

$$H_0: \mu_1 = \mu_2 = \mu_3 = \mu_4$$

$$H_A: \mu_1 \neq \mu_2 \neq \mu_3 \neq \mu_4$$

In order to test if there were any significant differences between the uninsured rates across the four regions among the three age groups, I would use the One Way ANOVA analysis on each of the age group tables to obtain the required F_{calc} and F_{crit} values to conduct the hypothesis test.

- *Rejection Rule: I will reject the null hypothesis if $F_{calc} > F_{crit}$*

IV: Findings/Observations

The results of my One Way ANOVA analyses for each of the age group tables is shown below:

Age Group: Under 19 years

ANOVA

<i>Source of Variation</i>	<i>SS</i>	<i>df</i>	<i>MS</i>	<i>F</i>	<i>P-value</i>	<i>F crit</i>
Between Groups	40.76057	3	13.58686	3.180752	0.032397	2.802355
Within Groups	200.7645	47	4.271586			
Total	241.5251	50				

Age Group: 19 - 64 years

ANOVA

<i>Source of Variation</i>	<i>SS</i>	<i>df</i>	<i>MS</i>	<i>F</i>	<i>P-value</i>	<i>F crit</i>
Between Groups	254.7689	3	84.92298	5.724864	0.002006	2.802355
Within Groups	697.2009	47	14.83406			
Total	951.9698	50				

Age Group: 65 years and older

ANOVA

<i>Source of Variation</i>	<i>SS</i>	<i>df</i>	<i>MS</i>	<i>F</i>	<i>P-value</i>	<i>F crit</i>
Between Groups	0.591075	3	0.197025	1.795595	0.160923	2.802355
Within Groups	5.157161	47	0.109727			
Total	5.748235	50				

Source: "Table HI05. Health Insurance Coverage Status and Type of Coverage by State and Age for All People: 2017" The United States Census Bureau, 2018

I then put the resulting F_{calc} and F_{crit} values from my One Way ANOVA analyses into a comprehensive hypothesis test table to make the results I gathered more organized and easier to understand.

Hypothesis Test Table

Age Group Test	F_{calc}	F_{crit}	$F_{calc} > F_{crit} ?$	Conclusion
Hypothesis Test 1: Under 19 years	3.1807523	2.8023552	Yes	Reject null
Hypothesis Test 2: 19 - 64 years	5.7248639	2.8023552	Yes	Reject null
Hypothesis Test 3: 65 years and over	1.7955946	2.8023552	No	Fail to reject

Source: "Table HI05. Health Insurance Coverage Status and Type of Coverage by State and Age for All People: 2017" The United States Census Bureau, 2018

So according to the hypothesis tests that I conducted, while there do exist differences in the average uninsured rates between U.S regions for both the under 19 years age group and the 19 – 64 years age group, there are not any significant differences between the average uninsured rates between the U.S regions for the age group that is 65 years and older. This initially had me somewhat surprised, but I could see how this was possible. The very fact that there was not a significant difference between uninsured rates between regions for the 65 years and older age group potentially revealed something that would be worth looking further into. What I am talking about in particular is an important aspect of the health insurance market that is relevant to economists and has its roots in economic theory: the issue of adverse selection. As has been stated earlier, the primary reason why many people opt to not go for health insurance is primarily because of cost. While there are other factors such as population characteristics and regional geography that do play a role in creating differences between regional uninsured rates, income differences between regions still likely plays an important role in influencing adverse selection in the health insurance market. This is an issue that has been particularly true for younger

nonelderly working age adults who often chose to not go for health insurance due to this age group generally considering themselves low health risk. When the costs are perceived to be higher than the benefits, this causes many people within this age group to drop out of the health insurance market. A somewhat current example of this taking place was when the tax penalty for the Affordable Care Act (ACA) was eliminated close to the end of 2018. The premium increase that resulted was a direct result of adverse selection, as healthier people dropped their coverage without the threat of a penalty resulting in a sicker group of people being left in the insurance pool. It would also appear that a somewhat similar story could be told for the under 19 year age group as well. I previously stated in my literature review that there are a number of families that earn too much to qualify for government provided health insurance, but do not earn enough to sufficiently afford health insurance of their own. For the same reasons that young nonelderly working age adults skip out on insurance coverage because they would consider themselves low health risk, a somewhat similar statement could be made for those under 19 as well. This brings us to the only age group that did not see a significant difference between its average regional uninsured rates. Considering that people over the age of 65 in all regions of the U.S regardless of income have access to health insurance coverage through Medicare likely explains why there is not a significant difference between average regional uninsured rates for this age group. It then appears certainly possible that if having sufficient income was not a necessary requirement for obtaining health insurance coverage for all of the age groups, then the regional differences between uninsured rates for all of the age groups could potentially decrease to the point where they would not be significant anymore.

There are various limitations that my analysis has that would affect my results. The first limitation being that my analysis is limited to the data that I can find online. If I cannot find

proper data from online sources to conduct analysis on, then my research essentially gets halted right there. Another limitation that I have is that if I were to actually try collecting my own data, I would not have the time or the resources to do so. Also considering how this is my first college statistics class and I am still very new to many statistical concepts, I myself would say that I am currently somewhat limited in how I can manipulate and analyze data.

V: Conclusion

So to sum up my findings, both the under 19 years age group and the 19 – 64 years age group do have different average uninsured rates across different regions. The 65 years and older age group on the hand, did not have significantly different average uninsured rates across the four U.S regions. While other factors do exist, both income and adverse selection in the health insurance market do likely play an important role in creating differences between uninsured rates for both different regions and age groups alike.

A lot of the statements that I made throughout this paper were based on the available data that I found online. While the sources that I did use can be considered credible and I do have confidence in my actual statistical calculations, I do feel that there could have been statements that I made in this paper that might not have been completely correct. The likely reason for this would be because I was unaware of other pieces of information that would have helped me have a better understanding on this topic and thus would allow my analysis on this topic to be more accurate.

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